

CORNERSTONE INTERNATIONAL COMMUNITY COLLEGE OF CANADA



CORNERSTONE
International Community College of Canada

DS 301 FINAL PROJECT

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GitHublink:

<https://github.com/Anna-Ikeda/Predicting-the-Success-of-Bank-Telemarketing-for-Selling-Long-term-Deposits.git>

1.Summary

This project focuses on predicting the success of bank telemarketing campaigns for selling long-term deposit products using supervised machine learning techniques. Telemarketing is a costly and resource-intensive strategy, and ineffective targeting can lead to significant financial losses. The primary objective of this study is **to develop and evaluate predictive models that can identify customers who are more likely to subscribe to long-term deposits**, thereby improving campaign efficiency and decision-making.

The project is based on reproducing and extending the methodology from the research paper *"Predicting the Success of Bank Telemarketing for Selling Long-term Deposits."* A publicly available dataset from a Portuguese bank was used, containing customer demographic, behavioral, and campaign-related features. The target variable is whether a customer subscribed to a long-term deposit, which is highly imbalanced.

To ensure reliable model evaluation, extensive data preprocessing was conducted, including one-hot encoding of categorical features, feature scaling, and stratified train-test splitting. Baseline models were first implemented to replicate the original study. The project was then extended through systematic hyperparameter tuning and the application of SMOTE to address class imbalance. Model performance was evaluated using Accuracy, Precision, Recall, F1-score, and ROC-AUC, with a particular emphasis on recall to minimize false negatives.

Overall, this project not only reproduces the findings of the original research but also enhances them by improving recall performance, incorporating additional classification models, and evaluating the robustness of the methodology under different modeling and resampling strategies.

2. Motivation :

The primary motivation of this research work is to use machine learning algorithms to optimize bank telemarketing campaigns in an effort to ensure the attraction of long-term deposits. Telemarketing campaigns are extremely costly; on the other hand, when these campaigns are directed at the wrong targets, the results include wasted time and money.

The project relies on a research paper entitled "Predicting the Success of Bank Telemarketing for Selling Long-term Deposits," which tackles this critical business issue by utilizing a standard dataset that can be recreated. A strong foundation for a baseline is thus created by recreating this scenario utilizing a respective Logistic Regression(LR), Decision Tree(DT) and Support Vector Machine(SVM) algorithm for this task. The study can then be further pursued utilizing other advanced algorithms such as K-Nearest Neighbors(KNN) and Neural Networks that could potentially be better suited for this specific classification task.

3. Research paper details

This study is based on the research paper titled ***“Predicting the Success of Bank Telemarketing for Selling Long-term Deposits: An Application of Machine Learning Algorithms” (2021)***, written by ***Premkumar Borugadda, Dr. Prabhakar Nandru, and Dr. Chendragiri Madhavaiah from Pondicherry University.***

This study is grounded in the context that globalization and technological advancement have enabled banks to reach customers worldwide, thereby expanding sales opportunities for their products and services. As part of this transformation, telemarketing has been introduced as a new sales approach. Telemarketing is a marketing method in which salespersons promote products to prospective customers over the phone and assess their willingness to purchase. It plays a dual role in both acquiring new customers and providing follow-up sales to existing customers.

In this research, publicly available data from telephone-based marketing campaigns conducted by a Portuguese bank are used to examine whether telemarketing is effective in promoting long-term bank deposits. **By applying multiple machine learning models, this study evaluates the effectiveness of telemarketing in influencing customers’ subscription decisions.** Machine learning techniques are particularly suitable for this problem because telemarketing outcomes depend on multiple customer attributes and complex interaction patterns that are difficult to capture using traditional rule-based or statistical approaches.

Previous studies applying machine learning to this topic are limited. Therefore, this research aims to contribute to the improvement of banking marketing strategies by identifying the conditions under which telemarketing is more likely to be successful.

4. Dataset details

Our dataset has the following 21 features:

Table 1. Features of the dataset

S. No	Features	Description of Features
1	Age	Age of the client
2	Job	Type of client's job
3	Marital	Client's marital status
4	Education	Highest education of the client
5	Default	Client credit
6	Housing	Housing loan
7	Loan	Personal loan
8	Contact	Type of contact communication with the client
9	Month	Last month of the year contracting to the client
10	Day of week	Last day of the week contracting to the client
11	Duration	Duration of client contact
12	Campaign	Number of contacts performed during the campaign and for this client
13	Plays	Number of days elapsed after the client's last visit
14	Previous	Number of contacts performed before this campaign and for this client
15	Poutcome	Outcome of the previous marketing campaign
16	Emp.var.rate	Employment variation rate
17	Cos.price.IDX	Consumer price index
18	Cons.conf.IDX	Consumer confidence index
19	Euribor3m	Euribor 3-month rate
20	Nr. employed	Number of employees
21	Label	Client subscription

Source: <https://archive.ics.uci.edu/ml/datasets/Bank+Marketing>

Our dataset consists of **4,120 rows and 21 features**. Among these features, **10 are numerical features** with data types *int64* and *float64*, and **11 are categorical features** representing customer demographic information, contact types (cellular or telephone), and campaign-related attributes. We use **20 features as descriptive (independent) variables**, while **“Label” is used as the target feature**, indicating whether a customer subscribed to a long-term deposit (Yes/No).

The dataset contains **imbalanced class distribution**, with fewer positive subscription cases compared to non-subscription cases. This characteristic makes it suitable for applying machine learning techniques and motivates the use of resampling methods such as **SMOTE**.

5.Data preprocessing and feature engineering

To prepare the Bank Marketing dataset for machine learning, we performed several essential steps:

- **Encoding:** Categorical variables were converted into numerical values using One-Hot Encoding (`pd.get_dummies`), ensuring the models could process non-numeric data. The target variable (`y`) was mapped to binary (1 for 'yes', 0 for 'no').
- **Splitting:** The data was split into a 90% training set and a 10% testing set. We used stratified sampling to ensure the ratio of subscribers to non-subscribers remained consistent across both sets.
- **Scaling:** We applied `StandardScaler` to normalize the feature values. This was a critical step for distance-based algorithms like SVM, KNN, and Neural Networks, which perform poorly when features vary widely in scale.

6.Steps reproduced from the paper

We successfully reproduced the core methodology of the original research paper:

- **Baseline Models:** We implemented Logistic Regression and Decision Tree classifiers as our baseline, mirroring the paper's approach.
- **Cross-Validation:** We utilized 10-fold Stratified Cross-Validation to validate model stability and ensure our performance metrics were reliable and not due to random chance.
- **Evaluation:** We assessed the models using the same key metrics as the paper: Accuracy, Precision, Recall, F1-Score, and AUC-ROC. This confirmed the paper's findings that while baseline models achieve high accuracy, they often struggle with Recall in imbalanced scenarios.

7.Contributions

Our contributions to this project are divided into three main tasks:

Performance Improvement Using Parameter Experimentation

The original research paper does not provide a detailed description of the methodology used for selecting model parameters. To address this limitation, we conducted additional experiments using multiple parameter configurations and applied systematic **hyperparameter tuning**. Furthermore, we incorporated **SMOTE** to handle class imbalance in the **Logistic Regression** and **Decision Tree** models. These approaches



extend the baseline experiments previously conducted without **SMOTE** or hyperparameter tuning in the **midterm stage**.

Incorporating Additional Classification Models

We implemented and evaluated additional classification models, including **SVM**, **KNN**, and **Neural Networks**. While SVM was introduced in the original research paper, it could not be implemented during the midterm project because the algorithm had not been covered in the course yet. In the final project, SVM was added to the experimentation process to demonstrate the application of previously learned algorithms and to extend the original methodology. Furthermore, we applied SMOTE to address class imbalance for these models and evaluated their performance before and after its application.

Evaluation of the Original Methodology on Different Datasets

Replication of processes from the midterm project, which ensured that results from this paper could be replicated by applying Logistic Regression and Decision Tree on the new dataset collected. One extension of the project was to add a SVM classifier to the model set and conduct analysis.

8. Significant improvements

Based on the contributions described above, this section highlights the significant improvements achieved in model performance and robustness.

Performance Improvement Using Parameter Experimentation

The following figures, derived from the Python code outputs, illustrate a slight performance improvement after applying hyperparameter tuning to the **Logistic Regression** and **Decision Tree** models. Additionally, a more substantial improvement is observed after incorporating **SMOTE**. These gains are primarily reflected in **recall**, although precision is negatively affected. Despite this trade-off, recall is prioritized in this study to **minimize false negatives**, based on the assumption that retaining potential customers who may subscribe to **long-term deposits** is more critical.

Logistic Regression:

- Early bird / After apply hyperparameter tuning

Classification Report:				
	precision	recall	f1-score	support
No	0.93	0.98	0.96	367
Yes	0.75	0.40	0.52	45
accuracy			0.92	412
macro avg	0.84	0.69	0.74	412
weighted avg	0.91	0.92	0.91	412

Logistic Regression Classification Report:				
	precision	recall	f1-score	support
0	0.93	0.98	0.95	367
1	0.73	0.36	0.48	45
accuracy			0.92	412
macro avg	0.83	0.67	0.72	412
weighted avg	0.90	0.92	0.90	412

- After apply **SMOTE**:

Logistic Regression Classification Report:				
	precision	recall	f1-score	support
0	0.98	0.87	0.92	367
1	0.46	0.89	0.61	45
accuracy			0.87	412
macro avg	0.72	0.88	0.77	412
weighted avg	0.93	0.87	0.89	412

Decision Tree:

- Early bird / After apply hyperparameter tuning

Classification Report:				
	precision	recall	f1-score	support
No	0.93	0.97	0.95	367
Yes	0.59	0.38	0.46	45
accuracy			0.90	412
macro avg	0.76	0.67	0.70	412
weighted avg	0.89	0.90	0.89	412

Decision Tree Classification Report:				
	precision	recall	f1-score	support
0	0.94	0.95	0.94	367
1	0.52	0.49	0.51	45
accuracy			0.90	412
macro avg	0.73	0.72	0.72	412
weighted avg	0.89	0.90	0.89	412

- After apply **SMOTE**:

Decision Tree Classification Report:				
	precision	recall	f1-score	support
0	0.97	0.87	0.92	367
1	0.43	0.78	0.55	45
accuracy			0.86	412
macro avg	0.70	0.82	0.73	412
weighted avg	0.91	0.86	0.88	412

Incorporating Additional Classification Models

In this project, **recall was prioritized** to minimize false negatives, while ROC-AUC was used to assess overall discrimination ability. Precision was also considered to ensure that the increase in recall did not lead to an excessive number of false positives. Based on these criteria, we will compare the model in the paper with the newly added model.

Applying additional classification models

These are result from applying additional classification models:

SVM

```
SVC Evaluation
-----
Accuracy   : 89.56%
Precision  : 53.12%
Recall     : 37.78%
F1 Score   : 44.16%
AUC Score  : 87.93%
Confusion Matrix:
[[352  15]
 [ 28  17]]

Classification Report:
              precision    recall  f1-score   support

      No         0.93      0.96      0.94       367
      Yes         0.53      0.38      0.44        45

   accuracy          0.90          412
  macro avg          0.73          412
 weighted avg          0.88          412
```

KNN


```
KNN Evaluation
-----
Accuracy      : 85.92%
Precision     : 32.43%
Recall        : 26.67%
F1 Score      : 29.27%
AUC Score     : 59.93%
Confusion Matrix:
[[342  25]
 [ 33  12]]

Classification Report:
              precision    recall  f1-score   support

      No         0.91      0.93      0.92       367
      Yes         0.32      0.27      0.29        45

   accuracy              0.86       412
  macro avg              0.62       412
 weighted avg              0.85       412
```

Neural Networks

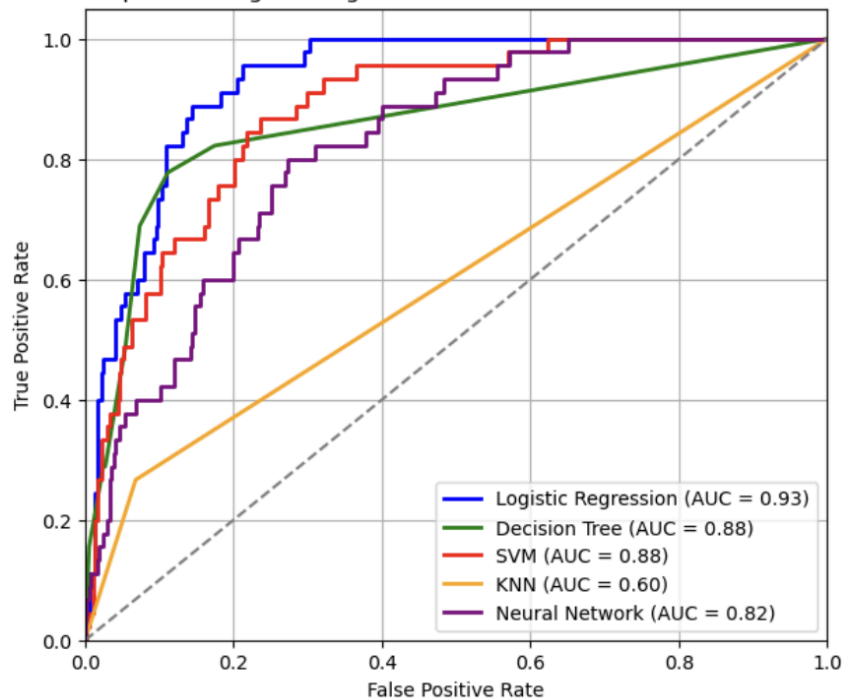
```
Neural Network Evaluation
-----
Accuracy      : 88.83%
Precision     : 48.39%
Recall        : 33.33%
F1 Score      : 39.47%
AUC Score     : 82.10%
Confusion Matrix:
[[351  16]
 [ 30  15]]

Classification Report:
              precision    recall  f1-score   support

      No         0.92      0.96      0.94       367
      Yes         0.48      0.33      0.39        45

   accuracy              0.89       412
  macro avg              0.70       412
 weighted avg              0.87       412
```

AUC-ROC Curve Comparison: Logistic Regression vs Decision Tree vs SVM vs KNN vs Neural Network



Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	91.99%	75.00%	40.00%	52.17%	92.98%
Decision Tree	89.56%	52.38%	48.89%	50.57%	85.76%
SVM	89.56%	53.12%	37.78%	44.16%	87.93%
KNN	85.92%	32.43%	26.67%	29.27%	59.93%
Neural Networks	88.83%	48.39%	33.33%	39.47%	82.10%

All models were optimized using hyperparameter tuning to ensure a fair and consistent comparison across classifiers.

Compared to the models discussed in the original paper (Logistic Regression and Decision Tree), the newly added models (SVM, KNN, and Neural Networks) did not demonstrate superior performance under the non-SMOTE setting. **Logistic Regression** remained the best-performing model in terms of accuracy and ROC-AUC, while Decision Tree achieved the highest recall.

Although all models were hyperparameter-tuned, the results without SMOTE indicate relatively low recall across all classifiers. This suggests that many potential customers were not correctly identified, likely due to class imbalance in the dataset. To address this limitation and improve minority-class detection, **SMOTE** was applied in subsequent experiments.

Applying SMOTE for additional models

To address class imbalance for these models and evaluated their performance, we applied SMOTE for SVM,KNN and Neural Networks.Here is the result:

SVM with SMOTE

```
SVC with SMOTE Evaluation
-----
Accuracy      : 84.95%
Precision     : 40.86%
Recall        : 84.44%
F1 Score      : 55.07%
AUC Score     : 92.98%
Confusion Matrix:
[[312  55]
 [  7 38]]

Classification Report:
              precision    recall  f1-score   support

      No         0.98        0.85        0.91        367
      Yes         0.41        0.84        0.55         45

   accuracy              0.85        412
  macro avg              0.69        0.85        0.73        412
weighted avg              0.92        0.85        0.87        412
```

KNN with SMOTE

KNN with SMOTE Evaluation

Accuracy : 79.61%
Precision : 33.04%
Recall : 84.44%
F1 Score : 47.50%
AUC Score : 82.47%

Confusion Matrix:

```
[[290  77]
 [  7  38]]
```

Classification Report:

	precision	recall	f1-score	support
No	0.98	0.79	0.87	367
Yes	0.33	0.84	0.47	45
accuracy			0.80	412
macro avg	0.65	0.82	0.67	412
weighted avg	0.91	0.80	0.83	412

Neural Networks with SMOTE

Neural Network with SMOTE Evaluation

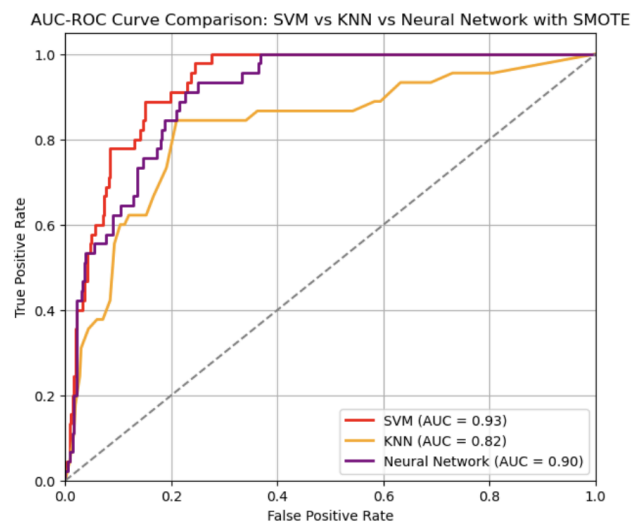
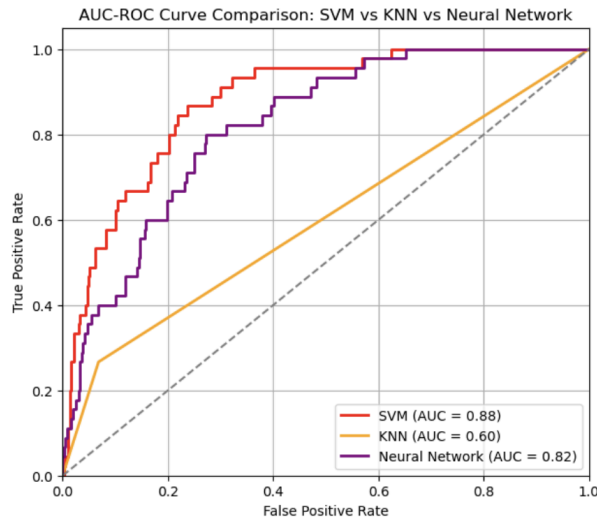
Accuracy : 88.35%
Precision : 47.17%
Recall : 55.56%
F1 Score : 51.02%
AUC Score : 90.48%

Confusion Matrix:

```
[[339  28]
 [ 20  25]]
```

Classification Report:

	precision	recall	f1-score	support
No	0.94	0.92	0.93	367
Yes	0.47	0.56	0.51	45
accuracy			0.88	412
macro avg	0.71	0.74	0.72	412
weighted avg	0.89	0.88	0.89	412

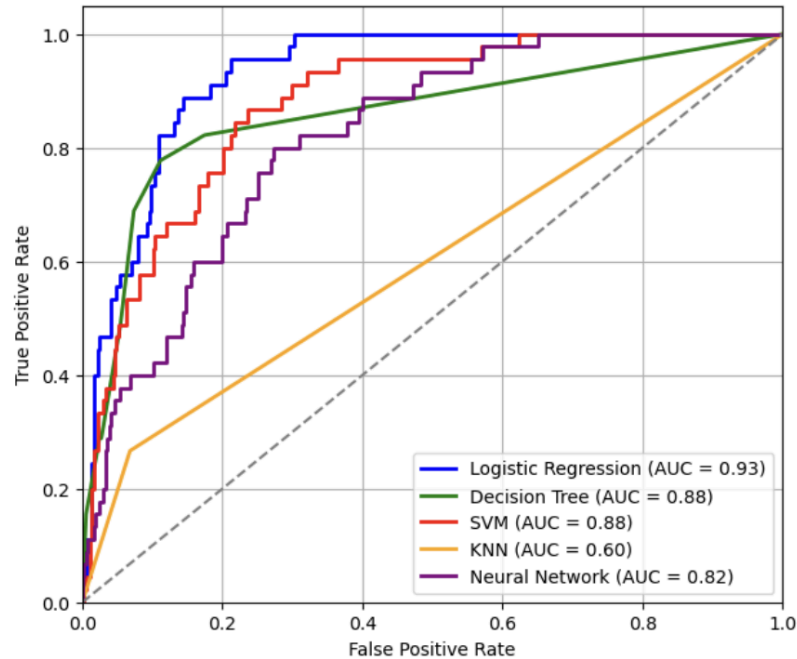


	Accuracy	Precision	Recall	F1-score	ROC-AUC
SVM	89.56%	53.12%	37.78%	44.16%	87.93%
SVM with SMOTE	84.95%	40.86%	84.44%	55.07%	92.98%
	Accuracy	Precision	Recall	F1-score	ROC-AUC
KNN	85.92%	32.43%	26.67%	29.27%	59.93%
KNN with SMOTE	79.61%	33.04%	84.44%	47.50%	82.47%
	Accuracy	Precision	Recall	F1-score	ROC-AUC
Neural Networks	88.83%	48.39%	33.33%	39.47%	82.10%
Neural Networks with SMOTE	88.35%	47.17%	55.56%	51.02%	90.48%

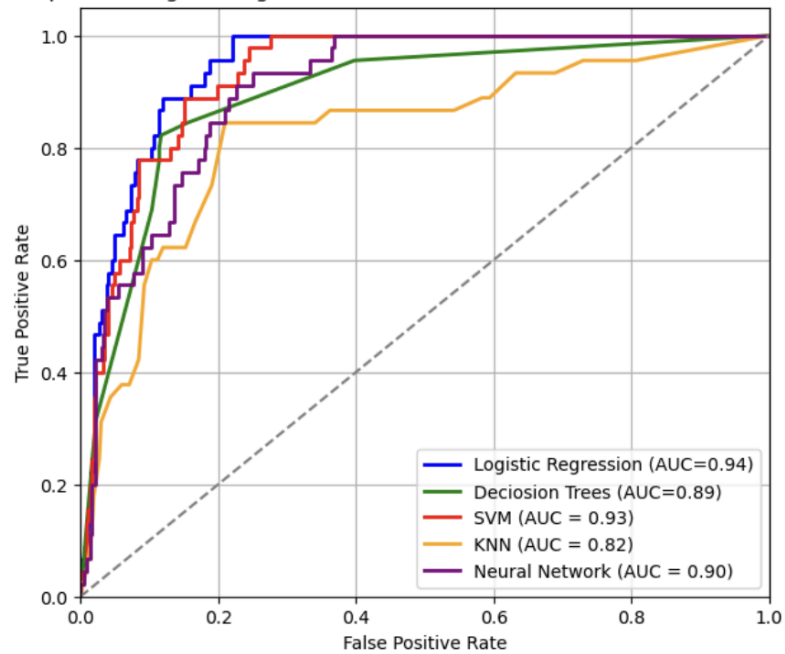
The comparison before and after applying SMOTE demonstrates a substantial improvement in recall across all additional models. In particular, SVM and KNN showed dramatic increases in recall, indicating enhanced detection of potential customers. Although this improvement results in lower accuracy and precision, the trade-off is acceptable because the primary objective of this study is to minimize false negatives and avoid missing potential customers. Therefore, **the application of SMOTE is considered effective and appropriate for this problem.**

Whether our contributions have improved the model performance or not

AUC-ROC Curve Comparison: Logistic Regression vs Decision Tree vs SVM vs KNN vs Neural Network



AUC-ROC Curve Comparison: Logistic Regression vs Decision Tree vs SVM vs KNN vs Neural Network with SMOTE



Model	SMOTE	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	No	91.99%	75.00%	40.00%	52.17%	92.98%
	Yes	87.14%	45.45%	88.89%	60.15%	94.28%
Decision Tree	No	89.56%	52.38%	48.89%	50.57%	85.76%
	Yes	87.62%	46.25%	82.22%	59.20%	89.33%
SVM	No	89.56%	53.12%	37.78%	44.16%	87.93%
	Yes	84.95%	40.86%	84.44%	55.07%	92.98%
KNN	No	85.92%	32.43%	26.67%	29.27%	59.93%
	Yes	79.61%	33.04%	84.44%	47.50%	82.47%
Neural Networks	No	88.83%	48.39%	33.33%	39.47%	82.10%
	Yes	88.35%	47.17%	55.56%	51.02%	90.48%

To identify the best-performing model, we compared Logistic Regression, Decision Tree, SVM, KNN, and Neural Networks under both SMOTE and non-SMOTE settings, using recall as the primary evaluation metric, supported by ROC-AUC and precision.

Based on the results, **Logistic Regression with SMOTE** was chosen as the best model. It achieved the highest recall (88.89%) and the highest ROC-AUC (94.28%), showing that it is very effective at identifying potential customers. Since the main goal of this project is to avoid missing customers who may subscribe to long-term deposits, this model offers the best overall balance among all models tested.

The additional models (SVM, KNN, and Neural Networks) did not perform better than the original models in the paper. However, the experiments provided useful insights. In particular, SVM achieved performance close to Logistic Regression in terms of ROC-AUC after applying SMOTE. This result shows that the original approach is strong and that other models can reach similar performance when class imbalance is properly handled. Therefore, our contributions improved the evaluation of the models by confirming and supporting the conclusions of the original study.

Evaluation of the Original Methodology on Different Datasets

Imbalance with SMOTE: To overcome imbalances in classes, SMOTE technique has been used where synthetic instances of the smaller class are introduced by SMOTE so that models refrain from outputting "No" predictions and work towards learning real customer features, thereby enhancing Recall, a key objective of identifying prospective consumers.

Optimizing SVM Training: Due to the presence of the entire dataset, the process of training the SVM became too time-intensive. We resorted to a stratified sample for the GridSearchCV for determining the parameters and then further trained the model on the whole dataset by implementing the parameters.

9.Challenges

Class Imbalance: The presence of an unequal distribution of classes for the target variables made it necessary for the models to favor one class over the other since it favored the "No" class to a large extent due to its dominance in the dataset. The use of SMOTE (Synthetic Minority Over-sampling Technique) helped to overcome this issue.

Computational Complexity of SVM: SVM is a computationally expensive model, which takes a very long time to train on the entire set of data. To address this issue, we formed a small set of data, which comprises only 10% of our training data. This was used in the GridSearchCV process to determine the optimal parameters. Later we used these optimal values to train SVM on our entire set of data. Regular, less computationally expensive methods may be less effective in performing similar tasks.

Lack of Domain Expertise: A key challenge of this project was the absence of input from a domain expert. Therefore, model decisions, such as prioritizing recall over precision, were based on assumptions rather than expert knowledge. We assumed that reducing false negatives is more important in telemarketing, as missing potential customers could lead to lost opportunities.

Team Collaboration: A key challenge was combining individual work into one unified codebase. Even though the same dataset was used, differences in interpretation and coding approaches required additional coordination. This emphasized the need for stronger communication within the team.

10.Conclusion and Future Scope

The results show that machine learning models can be effectively used to improve the targeting of bank telemarketing campaigns. Although baseline models achieved high accuracy, they performed poorly in terms of recall due to class imbalance.



Hyperparameter tuning led to small improvements, especially for Logistic Regression and Decision Tree models.

Applying SMOTE resulted in much larger improvements in recall across all models. While precision and accuracy decreased slightly, the increase in recall is more important for this business problem, as missing potential customers is more costly than contacting uninterested ones.

Among all evaluated models, Logistic Regression showed the most consistent and reliable performance, achieving the highest ROC-AUC score and strong recall. SVM also performed well after applying SMOTE, while KNN and Neural Networks were more sensitive to class imbalance but still benefited from resampling techniques.

Overall, this project successfully reproduces the original research and improves upon it by addressing class imbalance, testing additional models, and aligning evaluation metrics with business goals.

A couple of improvements can be explored in future work:

- **Alternative Imbalance Handling Methods**
Other resampling techniques such as ADASYN or cost-sensitive learning could be tested to better balance recall and precision.
- **Feature Selection**
Applying feature selection or dimensionality reduction techniques could improve model performance and reduce computational cost.